

Exercise 1 — Differential Index: 4 (or 3 to index-1)

DAE System

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = y \\ 0 = x_1 - r(t) \end{cases}$$

Variable identification

- **Differential variables:** x_1, x_2, x_3 (their first derivatives appear).
- **Algebraic variable:** y (Lagrange multiplier, no explicit differential equation initially).
- **Algebraic constraint:** $x_1 - r(t) = 0$.

Matrix form $\mathbf{E}(\mathbf{x}, t)\mathbf{x}' = \mathbf{G}(\mathbf{x}, t)$

Let $\mathbf{x} = [x_1, x_2, x_3, y]^\top$. Then:

$$\mathbf{E}^{(0)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{G}^{(0)} = \begin{bmatrix} x_2 \\ x_3 \\ y \\ r(t) - x_1 \end{bmatrix}.$$

The matrix $\mathbf{E}^{(0)}$ has rank 3 (singular).

Index reduction by successive differentiation

Level 0 (original):

$$\mathbf{E}^{(0)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{G}^{(0)} = \begin{bmatrix} x_2 \\ x_3 \\ y \\ r(t) - x_1 \end{bmatrix}.$$

Algebraic constraint: $g_0(x_1, t) = x_1 - r(t) = 0$.

Level 0

$\frac{\partial g_0}{\partial y} = 0 \implies$ the algebraic variable y does not appear $\implies$ not index 1.

Level 1: Differentiate g_0 :

$$\dot{g}_0 = \dot{x}_1 - r'(t) = x_2 - r'(t) = 0 \implies x_2 = r'(t).$$

New constraint (invariant): $g_1(x_2, t) = x_2 - r'(t) = 0$.

$$\mathbf{E}^{(1)} = \mathbf{E}^{(0)}, \quad \mathbf{G}^{(1)} = \begin{bmatrix} x_2 \\ x_3 \\ y \\ r'(t) - x_2 \end{bmatrix}.$$

Level 1

y is still absent $\implies$ not index 2.

Level 2: Differentiate g_1 :

$$\dot{g}_1 = \dot{x}_2 - r''(t) = x_3 - r''(t) = 0 \implies x_3 = r''(t).$$

New constraint (invariant): $g_2(x_3, t) = x_3 - r''(t) = 0$.

$$\mathbf{E}^{(2)} = \mathbf{E}^{(0)}, \quad \mathbf{G}^{(2)} = \begin{bmatrix} x_2 \\ x_3 \\ y \\ r''(t) - x_3 \end{bmatrix}.$$

Level 2

y is still absent $\implies$ not index 3.

Level 3: Differentiate g_2 :

$$\dot{g}_2 = \dot{x}_3 - r'''(t) = y - r'''(t) = 0 \implies y = r'''(t).$$

The algebraic variable y has appeared. The system is now index-1, but $\mathbf{E}^{(3)}$ is still singular:

$$\mathbf{E}^{(3)} = \mathbf{E}^{(0)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{G}^{(3)} = \begin{bmatrix} x_2 \\ x_3 \\ y \\ r'''(t) - y \end{bmatrix}.$$

Level 3

The algebraic variable y is uniquely determined. To obtain a non-singular $\mathbf{E}$ matrix (as requested by the exam sheet), we must differentiate once more.

Level 4 (Final ODE reduction): Differentiate $g_3(y, t) = y - r'''(t) = 0$:

$$\dot{g}_3 = \dot{y} - r^{(4)}(t) = 0 \implies \dot{y} = r^{(4)}(t).$$

Now the system has been fully converted into an ODE system where $\mathbf{E}^{(4)}$ is non-singular:

$$\mathbf{E}^{(4)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \mathbf{I}_4, \quad \mathbf{G}^{(4)} = \begin{bmatrix} x_2 \\ x_3 \\ y \\ r^{(4)}(t) \end{bmatrix}.$$

Differential Index = 4 (Steps to reach non-singular $\mathbf{E}^{(k)}$ form).

List of all invariants (algebraic constraints)

$$g_0(x_1, t) = x_1 - r(t) = 0$$

$$g_1(x_2, t) = x_2 - r'(t) = 0$$

$$g_2(x_3, t) = x_3 - r''(t) = 0$$

$$g_3(y, t) = y - r'''(t) = 0$$

Exercise 2 — Differential Index: 2

DAE System

$$\begin{cases} \dot{x} = y + \lambda \\ \dot{y} = -x \\ 0 = x + \sin t \end{cases}$$

Variable identification

- **Differential variables:** x, y
- **Algebraic variable:** λ (Lagrange multiplier).
- **Algebraic constraint:** $g(x, t) = x + \sin t = 0$.

DAE in matrix form $E\mathbf{x}' = \mathbf{G}$

Let $\mathbf{x} = [x, y, \lambda]^\top$, then:

$$\mathbf{E}^{(0)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{G}^{(0)} = \begin{bmatrix} y + \lambda \\ -x \\ -x - \sin t \end{bmatrix}.$$

Index reduction by successive differentiation

Level 0: Constraint $g_0(x, t) = x + \sin t = 0$.

Level 0

$$\frac{\partial g_0}{\partial \lambda} = 0 \implies \lambda \text{ does not appear} \implies \text{not index 1.}$$

Level 1: Differentiate g_0 :

$$\dot{g}_0 = \dot{x} + \cos t = (y + \lambda) + \cos t = 0 \implies y + \lambda + \cos t = 0.$$

The algebraic variable λ appears explicitly. The system is now index-1.

Differential index = 2 (Requires 2 differentiations to become a full ODE).

Reduced system (after one differentiation) — index-1

$$\begin{cases} \dot{x} = y + \lambda \\ \dot{y} = -x \\ 0 = y + \lambda + \cos t \end{cases}$$

In matrix form $\mathbf{E}^{(1)}\mathbf{x}' = \mathbf{G}^{(1)}$:

$$\mathbf{E}^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{G}^{(1)} = \begin{bmatrix} y + \lambda \\ -x \\ -y - \lambda - \cos t \end{bmatrix}.$$

List of all invariants (algebraic constraints)

$$\begin{aligned} g_0(x, t) &= x + \sin t = 0 \\ g_1(y, \lambda, t) &= y + \lambda + \cos t = 0 \end{aligned}$$

Projection step onto the first invariant $x + \sin t = 0$

Given an approximate solution $\tilde{\mathbf{x}} = [\tilde{x}, \tilde{y}, \tilde{\lambda}]^\top$, projecting ****only**** onto the first invariant means minimizing the distance subject exclusively to $g_0(x, t) = 0$. Since y and λ are unconstrained by g_0 , they remain unaltered:

$$\min_{x, y, \lambda} \|\mathbf{x} - \tilde{\mathbf{x}}\|^2 \quad \text{s.t.} \quad x + \sin t = 0$$

The explicit solution for the first box is:

$$\begin{cases} x = -\sin t \\ y = \tilde{y} \\ \lambda = \tilde{\lambda} \end{cases}$$

Note: Complete projection onto both invariants (for completeness)

If we project simultaneously onto both invariants $\mathcal{M} = \{\mathbf{x} : g_0 = 0, g_1 = 0\}$ via a Lagrangian formulation, we get:

$$\mathcal{L} = (x - \tilde{x})^2 + (y - \tilde{y})^2 + (\lambda - \tilde{\lambda})^2 + \mu_0(x + \sin t) + \mu_1(y + \lambda + \cos t)$$

Solving $\nabla \mathcal{L} = 0$ yields the joint orthogonal projection onto the manifold:

$$\begin{cases} x = -\sin t \\ y = \frac{\tilde{y} - \tilde{\lambda} - \cos t}{2} \\ \lambda = \frac{\tilde{\lambda} - \tilde{y} - \cos t}{2} \end{cases}$$